Saqya ساقية

The right amount of water, on your canal's schedule, in a text message.

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Live prototype: **eyad46.github.io/saqya**

THE PROBLEM

Egypt needs far more water than it has, and farming takes most of it

65 vs 88.5

billion m³ a year: renewable water against total demand

≈500 m³

per person per year, about half the 1,000 m³ line commonly used to define water scarcity

79%

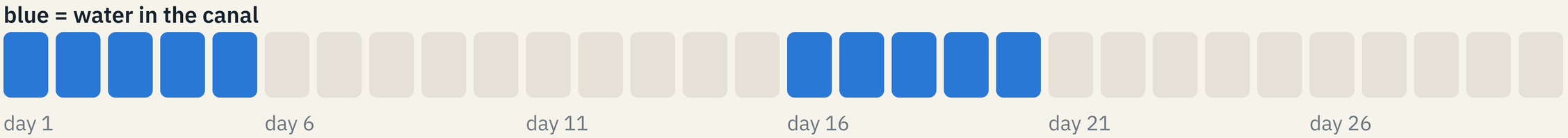
of all the freshwater Egypt withdraws goes to agriculture

Every cubic metre a farm does not waste is water for someone else.

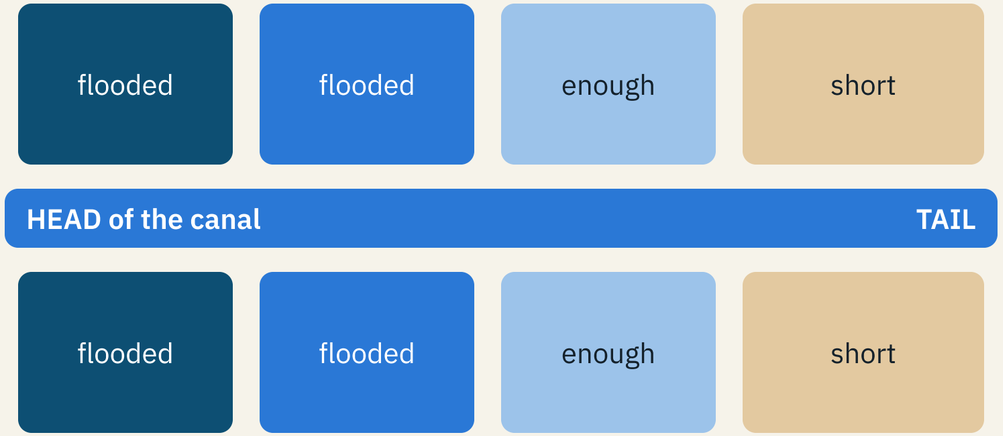
Sources: Minister of Water Resources and Irrigation Hani Sewilam, Ahram Online, 7 April 2026 (65.35 and 88.55 billion m³, about 500 m³ per person); World Bank indicator ER.H2O.FWAG.ZS (FAO AQUASTAT).

WHY WATER IS WASTED

In the Delta, a field only gets water on its canal's turn



- Winter rotation: about **5 days on, 10 days off**.
- Farmers who doubt the rotation tend to **over-irrigate**, especially at the head of the canal, and farmers at the tail get too little.
- Conventional wheat: **5 floods of about 435 m³ per feddan**, whatever the weather.



Sources: FAO, Rapid assessment study, irrigation and drainage in Egypt (IIIMP): "5 days on/10 days off in winter", head-end over-irrigation; Ghazouani et al. (2014), IWMI Research Report 160; El-Hadidi et al. (2016), Sakha field trial: 5 irrigations, 2,173 m³ per feddan.

WHAT EXISTS TODAY

Existing tools don't fit a farmer on a rotation canal

	SOIL SENSORS	SMARTPHONE APPS	SMS TIP SERVICES	SAQYA
Hardware in the field	**yes, per field**	**no**	**no**	**no**
Needs a smartphone and mobile data	often	**yes**	**no**	**no, plain SMS on any phone**
Plans around the canal's turns	**no**	**no, an interval between irrigations**	**no**	**yes, decides at every turn**
Tells the farmer what to do	a moisture reading	irrigation hours	weather and general tips	**skip, or pump for N minutes**

IWMI and FAO's IRWI app (2020) does the same water science for Delta farmers, but its developers said weak internet in the fields was their biggest obstacle.

Sources: "Innovative phone app helps Egypt farmers enhance water, soil productivity", Zawya, 2020; IWMI brief "Irrigation scheduling advice: Egypt", 2025; Vodafone Egypt, "Egyptian Farmers" service page (daily SMS with weather and farming tips, with the Ministry of Agriculture).

THE SOLUTION

One text message before every canal turn

Skip this turn. The soil holds enough water for your wheat until your next turn.

Light irrigation. Run your pump 4 h 30 min only. Don't flood the field.

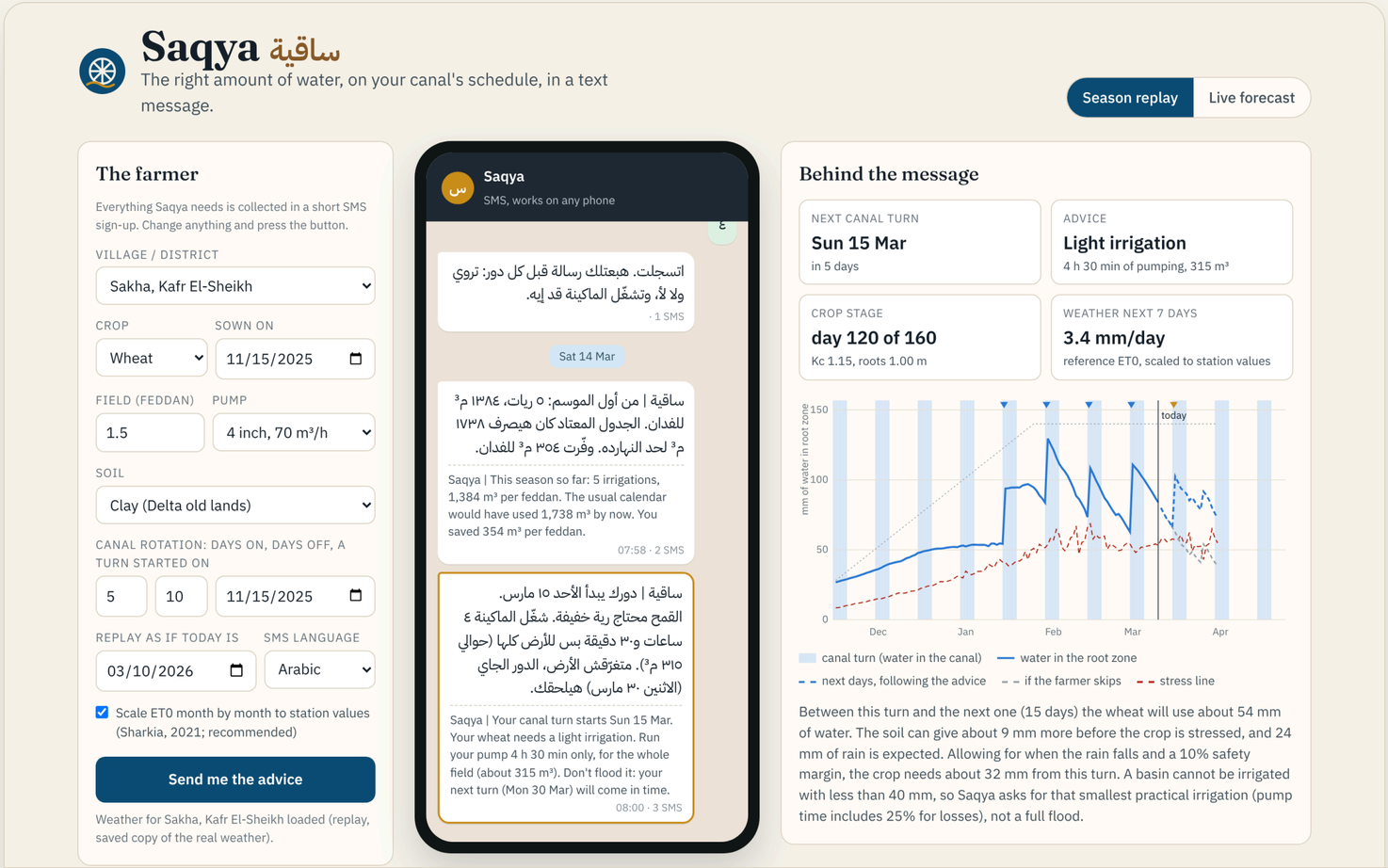
Full irrigation. Hot days ahead and your next turn is far.

any phone | Egyptian Arabic | no app, no internet, no sensors



WORKING PROTOTYPE

A working prototype: live forecast or season replay



Try it: **eyad46.github.io/saqya**

Any village, crop, canal schedule and pump. Shown here: a replay of 10 March 2026 with the real weather.

1. Weather, not guesswork

Every day Saqya pulls the free 16-day forecast for the farmer's

2. A soil water account

It keeps a running balance of the water left in the root zone, like

3. Planned around the canal

In the old lands water only comes on the canal's turn. Before

HOW SAQYA DECIDES

Standard science, one new question

1

Weather. Free 16-day forecast for the farmer's village (Open-Meteo): reference evapotranspiration and rain, scaled month by month to station-based values for Sharkia.

→

2

Water account. FAO-56 crop coefficients turn weather into crop water use by growth stage. Rain and irrigation in, crop use out, and the soil only holds so much.

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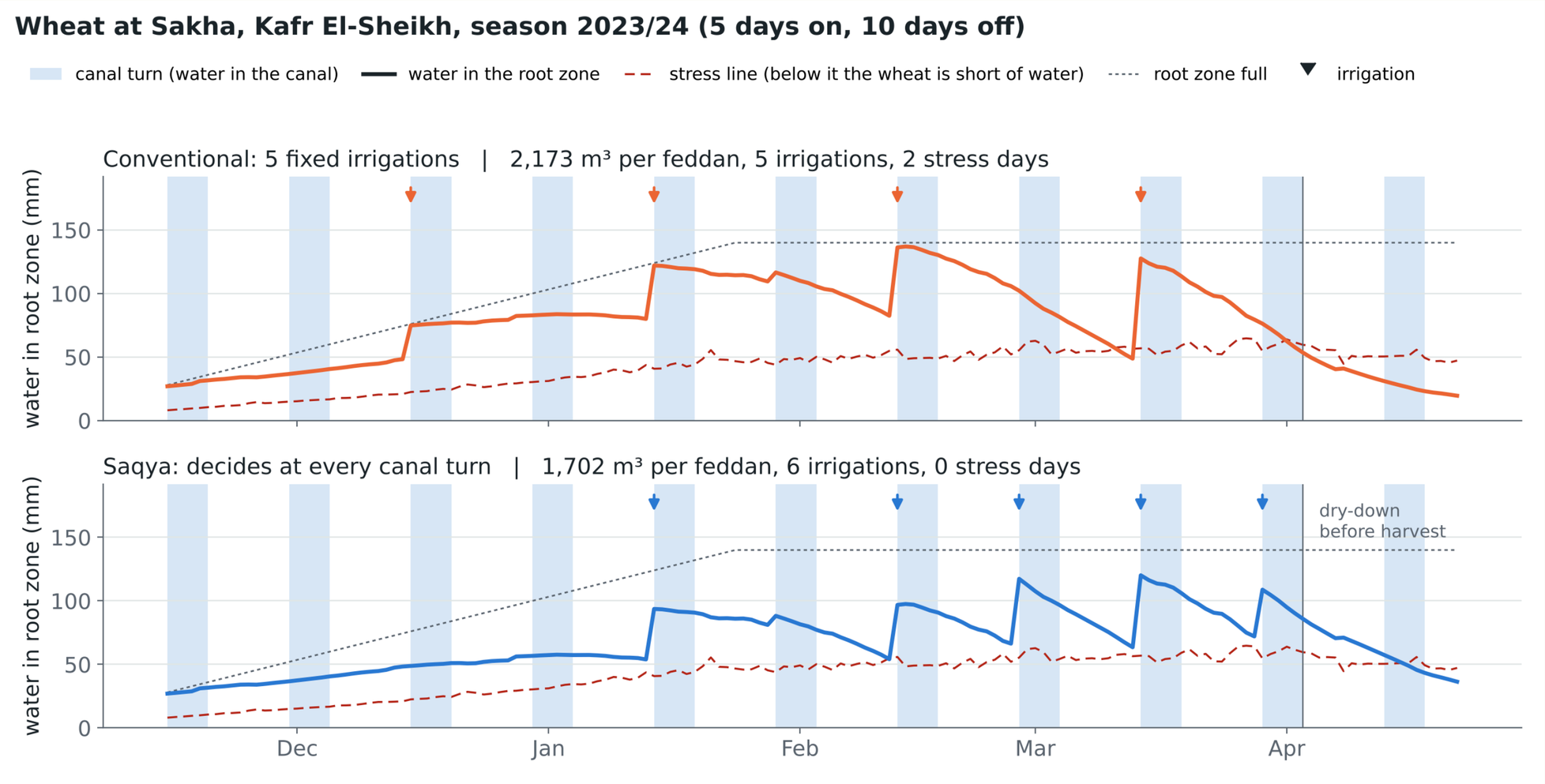
3

Canal-aware plan. At each turn: the least water that keeps the crop above its stress line until the next turn. Sent as pump minutes for the farmer's field.

Method: Allen et al. (1998), FAO Irrigation and Drainage Paper 56, Tables 11, 12, 19 and 22. Reanalysis ET0 is about 14 to 30% above station values from November to April, so it is scaled month by month to station-based ET0 for Sharkia (2021).

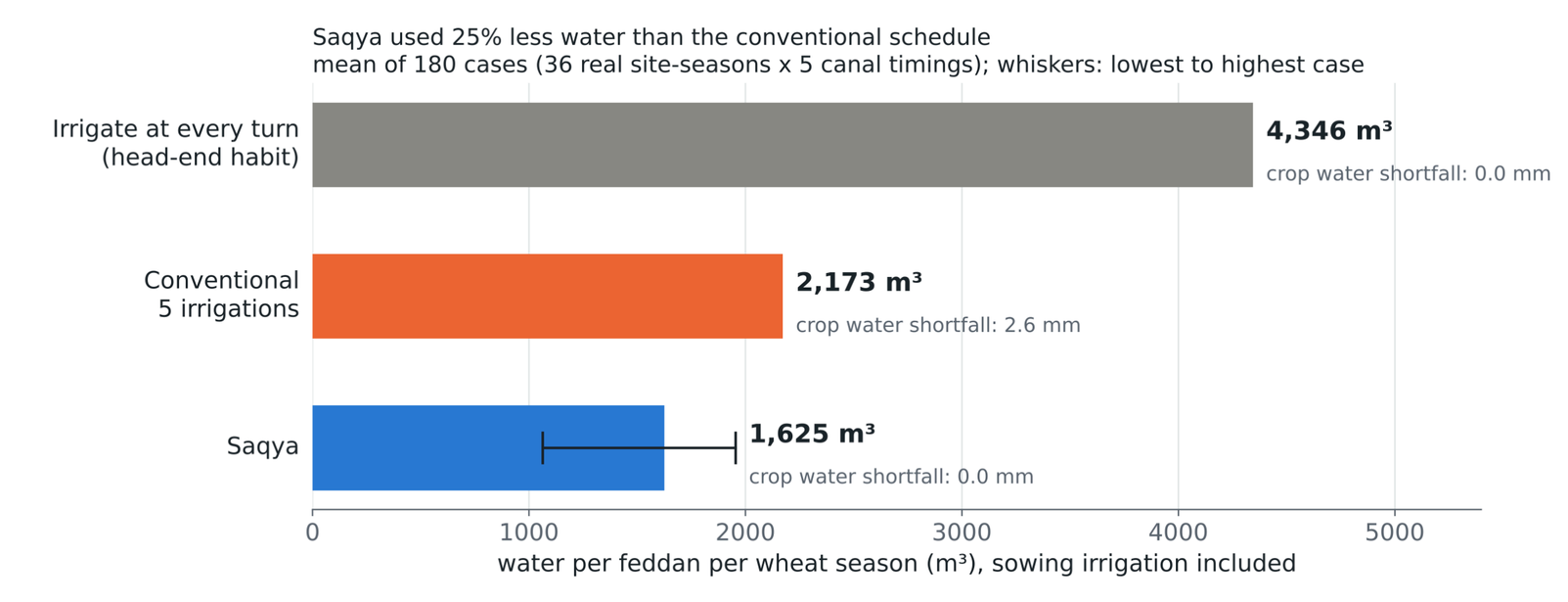
WHAT SAQYA ADDS

Same field, same weather, same canal



RESULTS

36 real seasons, 180 cases: 25% less water



548 m³
saved per feddan per wheat season, on average

180 / 180
cases where Saqya used less water

No extra stress
crop water shortfall 0.02 mm, against 2.6 mm for conventional

36 real site-seasons (6 Delta sites, winters 2019/20 to 2024/25), each with 5 canal timings. Saqya decided with a 7-day forecast with errors added and long-term averages after that. Conventional: 5 irrigations of 435 m³ per feddan (Sakha trial); irrigation days assumed (30, 60, 90, 120). Both at 80% application efficiency. With 22% higher crop water demand, the mean saving is 6% (Saqya used more in 56 of 180 cases).

IMPACT

What it could mean for Egypt

3.7 M

feddans of wheat in the 2025/26 season

55 M m³

less canal water drawn onto farms every winter if Saqya reached 100,000 of them (farm level)

≈8 h

less pumping per feddan per season with a 4-inch pump, so less diesel

Much of the water a farm wastes drains back and is reused downstream, so savings for the whole Nile system are smaller than on the farm. The clearest gains: water reaching farmers at the tail of the canal, and less diesel.

Sources: Ministry of Agriculture via Egyptian Cabinet (MEObserver, 14 April 2026); 548 m³ per feddan from the simulation; pump of 70 m³ per hour.

FEASIBILITY

Cheap to run, ready to pilot

About $1.00 to $1.60 per farmer per season. One advice SMS per turn plus a monthly summary, about 40 SMS parts in Arabic, at $0.025 to $0.04 per part with low-cost gateways (up to about $16 with the most expensive).

About $0.001 to $0.002 per cubic metre saved. It could also ride on an existing farmer SMS channel.

Free weather data, a public method, open-source code. No hardware to buy or repair.

1. Pilot (one season): 30 farmers on one canal in Kafr El-Sheikh with a water users' association; soil moisture checks on 10 fields to calibrate.

2. Grow: 1,000 farmers through extension workers; automated voice calls for farmers who do not read.

3. Scale: irrigation district offices send it to whole canals; more crops.

SMS prices to Egypt from a provider comparison of 8 gateways, $0.025 to $0.40 per SMS (sent.dm, prices checked January 2025). Cost per m³ assumes 1.5 feddan and 548 m³ saved per feddan, at low-cost rates.

LIMITS AND NEXT STEPS

Saving water can start with one text message

- The model assumes the soil under the young roots starts at field capacity (likely with a shallow water table). Half-depleted subsoil would cut the saving to about 11%.
- Savings need light irrigations: with a 60 mm smallest application the saving falls to 17%. Salinity is not modelled.
- The baseline comes from one field trial and the results are a simulation with real weather; the next step is a field pilot.

Thank you. شكراً

Live demo: **eyad46.github.io/saqya** · Code: **github.com/Eyad46/Eyad46.github.io/tree/main/saqya**

Eyad Selim, solo, Cairo. Outside help: AI assistants were used to help write code and edit text. All data sources and methods are cited.